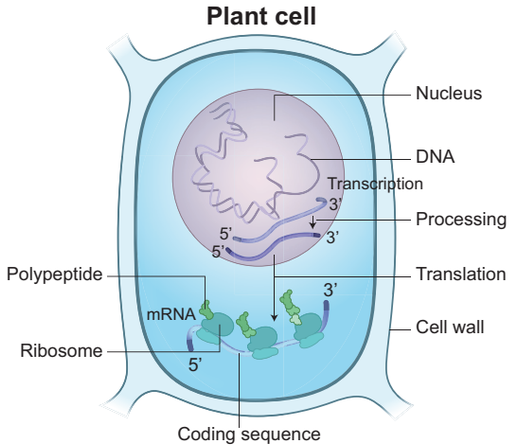


Figure: 3.20 Protein synthesis in plants

3.7.1 Transcription

Transcription is the process in which one strand of DNA acts as a template to generate mRNA with the bases complementary to the template strand. It is catalyzed by the enzymes called RNA polymerases.

Transcription and processing of RNA takes place in the **nucleus**, whereas the **translation** occurs in the ribosomes found in **cytoplasm**. In Eukaryotes, mRNA molecules are **monocistronic** with only one protein being derived from each mRNA.

The transcription begins with unwinding of DNA double helix and the hydrogen bonds are broken at the site of the gene being transcribed.

Template Strand / Non-Coding Strand / Antisense Strand

The strand of DNA which is oriented in 3' → 5' direction that serves as a template for the synthesis of mRNA is called template strand.

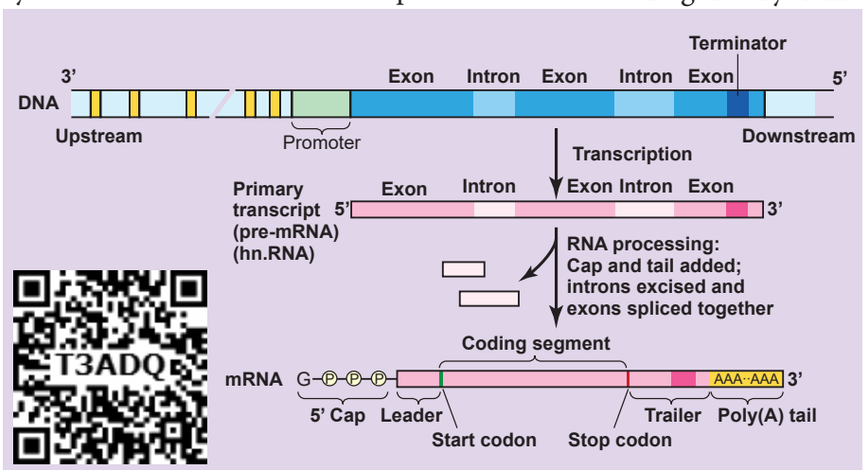


Figure: 3.21 Transcription

Coding strand / Non-template strand / Sense Strand

The other strand of DNA which is not transcribed is called the Coding Strand.

A specific sequence of DNA nucleotides called the **Promoter** is necessary for transcription to takes place. It consists of TATA box and transcription start site where transcription begins.

Termination sequences are the DNA sequences which tells when the RNA polymerase should stop producing RNA molecule.

Eukaryotic structural gene has 3 features in promoter

1. Regulatory elements
2. TATA box
3. A transcriptional start site

The transcription start site contains about 25 bp (basepairs) upstream, the sequence is TATAAT known as **TATA or Hogness box** which is present in core promoter. **General transcriptional factors** are the proteins which recognise base sequences of DNA and controls transcription. Some transcription factors bind directly to the promoter.

Some transcription factors recognize the regulatory elements and bind to them to increase the rate of transcription, others inhibits transcription.

To start the process of transcription the Regulatory elements help the RNA polymerase to recognize core promoter. The two categories of regulatory elements are

1. **Enhancer sequences** – they are DNA sequences (activating sequences) which help to influence transcription.
2. **Silencer sequence** – DNA sequences that inhibit transcription or decrease transcription.

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